Cairo University

Faculty of Computers and Artificial Intelligence

IS

Introduction to Database

Title: The Creative Media Production & Equipment Hub P20

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Section: S-33 && S-34

Version 1.0

Program: General

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1. System Description & Requirements

The Creative Media Production & Equipment Hub is a specialized management system designed for digital media agencies. It facilitates the coordination of large-scale video production projects by managing creative professionals, specialized studios, high-end cinema equipment, and production sessions.

Functional Requirements:

- **Professional Management:** Registering creative professionals (directors, cinematographers, editors) and mapping them to specific technical skills.
- **Studio Coordination:** Managing specialized production environments, including green-screen rooms and sound booths, across different wings of the headquarters.
- **Project Tracking:** Creating production projects with defined budgets, deadlines, and creative titles.
- **Session Scheduling:** Scheduling specific production dates, assigning them to studios, and designating teams of professionals.
- **Equipment Inventory:** Tracking cinema cameras and lighting kits, including unique serial numbers and recording the condition of gear upon return.

2. System Implementation Details

The application is developed using **C#** for the graphical user interface (GUI) and **Microsoft SQL Server** for the backend database.

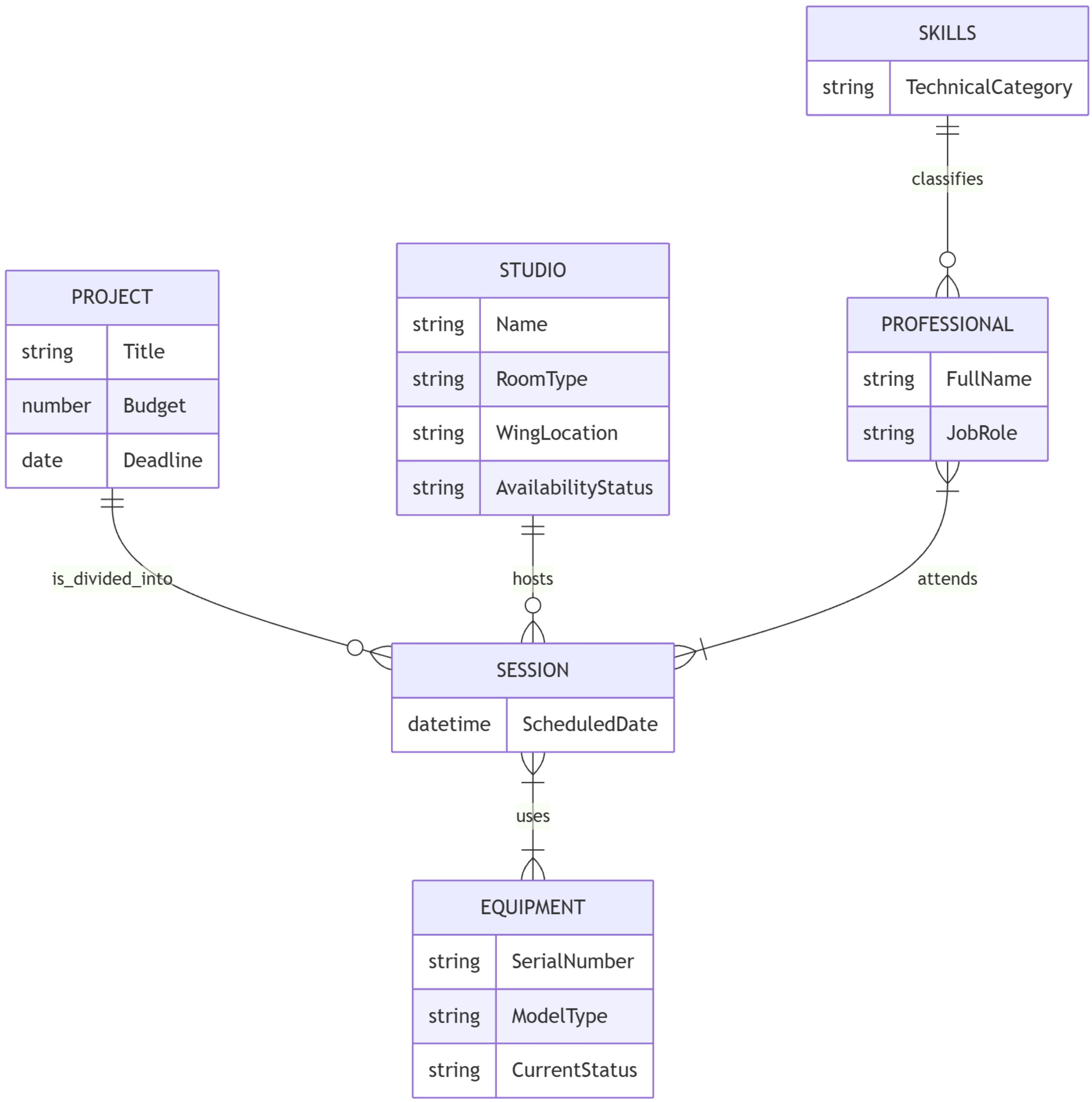
Key Technical Features:

- **Database Connectivity:** Utilizes ADO.NET (`SqlConnection`, `SqlCommand`) for direct communication with the database.
- **No ORM Policy:** All SQL statements (Insert, Update, Delete, Select) were written manually in the application backend to ensure optimized data handling and compliance with project constraints.
- **CRUD Operations:** The system implements functional data entry for all primary entities, allowing for the addition, modification, and removal of equipment, professionals, and project records.
- **Advanced Inquiries:** The system features complex SQL join queries to retrieve multi-table data, such as tracking equipment utilized across specific projects and analyzing professional contributions.

3. Conceptual Data Model (ERD)

Note: [Insert Image of ConceptualDataModel.cdm here]

The Conceptual ERD defines the high-level entities of the system. It includes at least 5 core entities: **PROJECT, SESSION, STUDIO, PROFESSIONAL,** and **EQUIPMENT**. Relationships are established to ensure a many-to-many relationship exists between professionals and sessions (represented by the **WORK_ON** entity), and between equipment and sessions (represented by **SESSION_EQUIP**).



4. Physical Data Model (PDM)

Note: [Insert Image of PhysicalDataModel.pdm here]

The Physical Data Model translates the conceptual design into a schema compatible with MS SQL Server. It defines:

- Primary Keys (e.g., `EQ_ID`, `PROFESSIONAL_ID`) and Foreign Keys to maintain referential integrity.
- Specific data types for all attributes (e.g., `DATETIME` for deadlines, `DECIMAL` for budgets).
- Nullability constraints and table relationships to ensure data consistency across the equipment hub.